

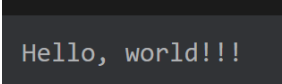
Aim: Basic programming in Kotlin.

1. Basic print code.

CODE:

```
fun main() {  
    println("Hello, world!!!")  
}
```

OUTPUT:



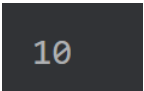
```
Hello, world!!!
```

2. Variable declaration Read-Only and Mutable.

CODE:

```
fun main() {  
    val a = 10  
    println(a)  
}
```

OUTPUT:

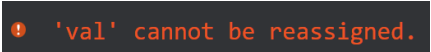


```
10
```

CODE:

```
fun main() {  
    val a = 10  
    a = 20  
    println(a)  
}
```

OUTPUT:



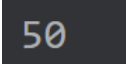
```
❗ 'val' cannot be reassigned.
```

3. Basic Operations.

CODE:

```
fun main() {  
    var a = 10  
    a = 50  
    println(a)  
}
```

OUTPUT:



```
50
```

CODE:

```
fun main() {  
    var a = 10  
    println(a)  
    a = 8  
    println(a)  
    a = a + 3  
    println(a)  
    a += 7
```

```
println(a)
a -= 3
println(a)
a *= 2
println(a)
a /= 3
println(a)
a %= 2
println(a)
}
```

OUTPUT:

```
10
8
11
18
15
30
10
0
```

4.Variable initialization with datatypes.**CODE:**

```
fun main() {
    val a:Double
    a = 90.5
    println(a)
}
```

OUTPUT:

```
90.5
```

CODE:

```
fun main() {
    val a:String = "Atomic"
    println("Mr. $a")
}
```

OUTPUT:

```
Mr. Atomic
```

5.List datatypes.**CODE:**

```
fun main() {
    val r = listOf("Atomic", "Ghost", "Dexter", "Z_17")
    println(r)
}
```

OUTPUT:

```
[Atomic, Ghost, Dexter, Z_17]
```

CODE:

```
fun main() {  
    val m = mutableListOf("Atomic", "Ghost", "Dexter", "Z_17")  
    println(m)  
}
```

OUTPUT:

```
[Atomic, Ghost, Dexter, Z_17]
```

CODE:

```
fun main() {  
    val m = mutableListOf("Atomic", "Ghost", "Dexter", "Z_17")  
    println(m)  
    m.add("Shadow46")  
    println(m)  
    m.remove("Dexter")  
    println(m)  
    m[1] = "Dex"  
    println(m)  
}
```

OUTPUT:

```
[Atomic, Ghost, Dexter, Z_17]  
[Atomic, Ghost, Dexter, Z_17, Shadow46]  
[Atomic, Ghost, Z_17, Shadow46]  
[Atomic, Dex, Z_17, Shadow46]
```

6.SET datatypes.**CODE:**

```
fun main() {  
    val s = setOf("Atomic", "Ghost", "Dexter", "Z_17")  
    println(s)  
}
```

OUTPUT:

```
[Atomic, Ghost, Dexter, Z_17]
```

CODE:

```
fun main() {  
    val s = mutableSetOf("Atomic", "Ghost", "Dexter", "Z_17")  
    println(s)  
    s.add("Shadow46")  
    println(s)  
    s.remove("Dexter")  
    println(s)  
    s.add("Dexter")  
    println(s)  
}
```

OUTPUT:

```
[Atomic, Ghost, Dexter, Z_17]  
[Atomic, Ghost, Dexter, Z_17, Shadow46]  
[Atomic, Ghost, Z_17, Shadow46]  
[Atomic, Ghost, Z_17, Shadow46, Dexter]
```

7.SET-Count.

CODE:

```
fun main() {  
    val s = setOf("Atomic", "Ghost", "Dexter", "Z_17", "Z_17")  
    println("This set has ${s.count()} items")  
}
```

OUTPUT:

```
This set has 4 items
```

CODE:

```
fun main() {  
    val s = setOf("Atomic", "Ghost", "Dexter", "Z_17", "Z_17")  
    println("Atomic" in s)  
}
```

OUTPUT:

```
true
```